

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (IT)  
March-April 2018

IT242/IT216 Java Programming

Date: 28.03.2018, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

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**Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## **SECTION - I**

**Q - 1**

- (a) Explain various Object Oriented Programming Concepts in detail. [04]
- (b) What is Code Reusability (Inheritance)? Explain multilevel inheritance in detail. [03]

**Q - 2**

- (a) Explain various data type available in Java with suitable example. [03]
- (b) How final keyword modifies the role of class, method and variable? [04]
- (c) Write a program to print "Hello Charusat" and explain the Java Programming Life Cycle in detail. [07]

## **OR**

**Q - 2**

- (a) How Java is differ from C++? Explain with example. [03]
- (b) Explain the different methods of String class to manipulate characters from the string. [04]
- (c) Write a program to enter your FirstName and LastName through command line and try to display full name. (e.g. First Name: ABC, Last Name: DEF, Output: ABC DEF) [07]

**Q - 3**

- (a) What is run time polymorphism? Explain with example. [03]
- (b) Give the demonstration of the following terms: [04]
  - 1. super keyword
  - 2. constructor overloading
- (c) What is interface? How interface is differ from class? Explain with example. [07]

## **OR**

**Q - 3**

- (a) List the types of logical operator. Explain any one of them briefly. [03]
- (b) Explain the following terms: [04]
  - 1. this keyword
  - 2. garbage collector
- (c) What is an abstract class? Explain with example. [07]

## **SECTION – II**

**Q - 4**

- (a) Explain different methods available in Applet life cycle. [05]
- (b) Differentiate BufferedReader and Scanner class. [02]

**Q - 5**

- (a) How GUI based application is differ from Console based application? Explain <APPLET> tag. [07]
- (b) How to handle an exception occurred in your program. Explain basic keywords associated with Exception Handling. [07]

**OR**

**Q – 5**

- (a) What is Thread? Narrate the difference between multithreading and multiprocessing in detail. [07]
- (b) Write a program in Java to demonstrate ArithmeticException. [07]

**Q - 6**

- (a) Why synchronization is required in multithreading? [07]
- (b) Write a program to demonstrate the use of ActionListener in Java. [07]

**OR**

**Q - 6**

- (a) Explain following classes briefly: [07]
  - 1. Adapter Class
  - 2. Thread Class
- (b) Write down the steps for executing an applet program. [07]

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